

Non-unimodular $\hat{1}$ SUGRA Bases: target = (-1)

$\hat{1}$ external: $C^2 = -1$, $b_0 \cdot C = -1$.

Total: 24 bases (from 24 distinct LST bases, $T_{\text{LST}} \in [0, 30]$).

Exact det = 0: 24, fractional T_{crit} : 0. Filtered: $\Delta_{\text{crit}} \leq 0$.

$\hat{1}$ = attached external.

Intersection matrix convention: Each matrix $(\cdot)_{b_0}$ is the intersection form of the SUGRA base, obtained by extending the LST intersection matrix by $b_0^2 = 9 - T_{\text{crit}}$ and then blowing down all (-1) -curves until no further blowdown is possible. The resulting matrix has signature $(1, T, n_0)$ where n_0 counts null directions (if any).

T	Count
2	1
3	2
4	2
5	2
6	3
7	3
8	3
9	4
10	4

T_{crit}	Count
10	17
11	3
12	2
13	2

$ \det $	Count
4	24

$T = 2$ (1 bases)

Base: 11

$$\begin{aligned} 11 + \hat{1} \quad \Delta = -242, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 2, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 2, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0} \end{aligned}$$

$T = 3$ (2 bases)

Base: 121

$$\begin{aligned} 121 + \hat{1} \quad \Delta = -213, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 3, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 3, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \end{aligned}$$

Base: 212

$$\begin{aligned} \hat{1} 212 \quad \Delta = -213, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 3, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 3, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \end{aligned}$$

$T = 4$ (2 bases)

Base: 1221

$$\begin{aligned} 1221 + \hat{1} \quad \Delta = -184, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 4, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \end{aligned}$$

Base: $\overset{2}{122}$

$$\begin{aligned} \overset{2}{221} \hat{1} \quad \Delta = -184, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 4, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \end{aligned}$$

$T = 5$ (2 bases)

Base: 12221

$$\begin{aligned} 12221 + \hat{1} \quad \Delta = -155, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 5, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

Base: $\overset{2}{1222}$

$$\begin{aligned} \overset{2}{2221} \hat{1} \quad \Delta = -155, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 5, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$T = 6$ (3 bases)

Base: 122221

$$\begin{aligned} 122221 + \hat{1} \quad \Delta = -126, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 6, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0} \end{aligned}$$

Base: $\overset{2}{12222}$

$$\begin{aligned}
& \overset{2}{22221}\hat{1} \quad \Delta = -126, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0) \\
& T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1) \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

Base: $\overset{31}{1232}$

$$\begin{aligned}
& \overset{31}{2321}\hat{1} \quad \Delta = -145, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0) \\
& T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1) \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$T = 7$ **(3 bases)**

Base: 1222221

$$\begin{aligned}
& 1222221 + \hat{1} \quad \Delta = -97, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0) \\
& T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1) \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

Base: $\overset{2}{122222}$

$$\begin{aligned}
& \overset{2}{222221}\hat{1} \quad \Delta = -97, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0) \\
& T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1) \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

Base: $\overset{2}{231321}$

$$\begin{aligned}
& \overset{2}{231321}\hat{1} \quad \Delta = -117, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0) \\
& T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1) \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$T = 8$ **(3 bases)**

Base: 12222221

$$\begin{aligned}
& 12222221 + \hat{1} \quad \Delta = -68, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0) \\
& T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1) \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 6 \\ 2 & 6 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

Base: $\overset{2}{1222222}$

$$\begin{aligned}
& \overset{2}{2222221}\hat{1} \quad \Delta = -68, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0) \\
& T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1) \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 6 \\ 2 & 6 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

Base: $\overset{2}{2313221}$

$$\begin{aligned}
& \overset{2}{2313221}\hat{1} \quad \Delta = -88, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0) \\
& T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1) \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

$T = 9$ **(4 bases)**

Base: 122222221

$$\begin{aligned}
& 122222221 + \hat{1} \quad \Delta = -39, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0) \\
& T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1) \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 7 \\ 2 & 7 & 7 \end{pmatrix}_{b_0}
\end{aligned}$$

Base: 1222222²2

$$2^{22222221} \hat{1} \quad \Delta = -39, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 9, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 7 \\ 2 & 7 & 7 \end{pmatrix}_{b_0}$$

Base: 1231413²2

$$2^{\hat{1}123141} 3^2 \quad \Delta = -87, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 9, 0) \\ T_{\text{crit}} = 12, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

Base: 31¹²³¹5¹³

$$31^{\hat{1}1231} 5^{13} \quad \Delta = -116, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 9, 0) \\ T_{\text{crit}} = 13, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$T = 10$ **(4 bases)**

Base: 1222222221

$$1222222221 + \hat{1} \quad \Delta = -10, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 10, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 8 \\ 2 & 8 & 8 \end{pmatrix}_{b_0}$$

Base: 1222222²22

$$2^{22222221} \hat{1} \quad \Delta = -10, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 10, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 8 \\ 2 & 8 & 8 \end{pmatrix}_{b_0}$$

Base: 12231413²2

$$2^{\hat{1}1223141} 3^2 \quad \Delta = -58, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 10, 0) \\ T_{\text{crit}} = 12, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 6 \\ 2 & 6 & 6 \end{pmatrix}_{b_0}$$

Base: 31¹²²³¹5¹³

$$31^{\hat{1}12231} 5^{13} \quad \Delta = -87, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 10, 0) \\ T_{\text{crit}} = 13, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$